Project Design Phase-II

Data Flow Diagram & User Stories

Date	14 July 2026
Project Name	Food Ordering Behaviour and Customer Trends
Team Member	Bhoomi Jain
Maximum Marks	4 Marks

Data Flow Diagrams Description

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD LEVEL 0 - CONTEXT DIAGRAM

The **Food Ordering Behaviour & Customer Trends Analytics System (0.0)** ingests Raw Order Data from the **Food Ordering Dataset CSV (D1, ~50,000 records)**. The **Restaurant Manager** provides User Filter Selections and interacts with the Interactive Dashboard & Business Insights. The **Business Analyst** inputs User Filter Selections and receives the Tableau Story & Business Reports.

DFD LEVEL 1 - PROCESS BREAKDOWN

The granular flow follows a step-by-step pipeline:

1. **1.0 Data Cleaning**: Ingests raw data to handle missing or corrupt records, saving to **D2: Cleansed Raw Dataset**.
2. **2.0 Data Preparation**: Transforms data into forms, categories, and aggregates, writing to **D3: Prepared Analysis Dataset**.
3. **3.0 Interactive Dashboard Development**: Generates core component metrics (KPIs, Cuisines, Systems, Ratings, Delivery, Meal Type, and Trends), saving into **D4: Repository Components**.
4. **4.0 Tableau Story Development**: Compiles views into an analytical story line.
5. **5.0 Flask Web Application**: Hosts the platform, managing frontend/backend integration and final serving to **Dashboard Users** (Restaurant Managers, Business Analysts, Decision Makers).

User Stories

User Type	Epic	No.	User Story / Task	Acceptance Criteria	Priority	Release
Restaurant Manager	KPI Dashboard	USN-1	As a Restaurant Manager, I want to view key KPIs such as total orders, revenue, average rating, and average delivery time so that I can	KPI cards display accurate, up-to-date totals for orders, revenue, rating, and	High	Sprint-1

User Type	Epic	No.	User Story / Task	Acceptance Criteria	Priority	Release
			quickly assess overall business performance.	delivery time when dashboard loads.		
	Cuisine Analysis	USN-2	As a Restaurant Manager, I want to analyze cuisine popularity so that I can optimize menu planning.	Dashboard displays order count and revenue by cuisine type, viewable as a sortable chart.	High	Sprint-1
	Meal Type Analysis	USN-3	As a Restaurant Manager, I want to see order distribution across meal types (breakfast, lunch, dinner, snacks) so that I can plan staffing and inventory accordingly.	Meal type chart updates correctly whenever a filter is applied.	Medium	Sprint-2
	Restaurant Segment	USN-4	As a Restaurant Manager, I want to compare performance across restaurant segments so that I can identify high- and low-performing categories.	Segment-wise chart shows revenue and order share for each restaurant segment.	Medium	Sprint-2
	Interactive Filters	USN-5	As a Restaurant Manager, I want to filter dashboard data by date range and city so that I can focus on a specific period or location.	Selecting a filter updates all visible charts and KPIs without a page reload.	High	Sprint-1
Business Analyst	Customer Rating	USN-6	As a Business Analyst, I want to compare customer ratings across cuisines and restaurants so that I can identify service improvement opportunities.	Rating comparison view highlights restaurants or cuisines that fall below the average rating.	High	Sprint-2
	Delivery Time	USN-7	As a Business Analyst, I want to analyze average delivery time trends so that I can flag delays affecting customer satisfaction.	Delivery time chart shows a trend line and highlights delays beyond an acceptable threshold.	High	Sprint-2
	City-wise Analysis	USN-8	As a Business Analyst, I want to view order volume and revenue by city so that I can identify regional demand patterns.	City-wise chart reflects order counts and revenue that match the underlying dataset.	Medium	Sprint-3
	Customer Age Group	USN-9	As a Business Analyst, I want to see ordering behaviour by customer age group so that I can understand demographic preferences.	Age-group breakdown chart can be filtered by cuisine and meal type.	Low	Sprint-3
	Revenue Analysis	USN-10	As a Business Analyst, I want to track revenue trends over time so that I can identify seasonal patterns in ordering behaviour.	Revenue trend chart displays weekly and monthly totals with correct aggregation.	High	Sprint-1
	Tableau Story	USN-11	As a Business Analyst, I want to walk through a guided Tableau Story summarizing key findings so that I can present insights without rebuilding charts from scratch.	Story points navigate in sequence and each point displays its correct supporting visualization.	Medium	Sprint-3
Decision Maker	KPI Dashboard	USN-12	As a Decision Maker, I want to monitor overall KPI metrics at a glance so that I can make informed business decisions quickly.	KPI summary section is visible without scrolling when the dashboard loads.	High	Sprint-1
		USN-13			Medium	Sprint-4

User Type	Epic	No.	User Story / Task	Acceptance Criteria	Priority	Release
	Business Reports		As a Decision Maker, I want to export a summary of dashboard insights so that I can share findings with stakeholders outside the dashboard.	Export option produces a report or image reflecting the current filtered view.		
	Revenue Analysis	USN-14	As a Decision Maker, I want to compare revenue performance across cuisines and cities so that I can prioritize investment decisions.	Comparative revenue view allows two or more categories to be selected side by side.	Medium	Sprint-3
	Restaurant Segment	USN-15	As a Decision Maker, I want to identify the highest-performing restaurant segments so that I can decide where to focus business expansion.	Segment ranking table sorts restaurants by revenue or order volume in descending order.	Low	Sprint-4
Dashboard User	Flask Website	USN-16	As a Dashboard User, I want to access the embedded Tableau Dashboard and Story through a simple web page so that I can view insights without installing any software.	Flask web page loads the embedded dashboard and story correctly in common browsers.	High	Sprint-1
	Dashboard Visualization	USN-17	As a Dashboard User, I want the dashboard layout to be clear and easy to navigate so that I can find the information I need without confusion.	Navigation between dashboard sections (KPI, Cuisine, Revenue, Rating, etc.) is available from a single visible menu.	Medium	Sprint-2
	Interactive Filters	USN-18	As a Dashboard User, I want to reset all filters with a single action so that I can quickly return to the default dashboard view.	A Reset Filters control clears all active filters and restores the default view.	Low	Sprint-2